

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & 0 & M_z \end{bmatrix}$$

$$M_z = M_{pz}^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} M_{xx} & 0 & 0 \\ 0 & -M_{xx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xx} = M_{dv}^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & M_{xxz} \\ 0 & 0 & M_{zzz} \\ 0 & M_{xxz} & 0 \\ M_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxz} = -M_{faz}^{(1)} + M_{pz}^{(1)}$$

$$M_{zzz} = 2M_{faz}^{(1)} + 3M_{pz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & M_{xxz} \\ 0 & 0 & 0 \\ 0 & -\frac{M_{xxz}}{2} & 0 \\ -\frac{M_{xxz}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxz} = 2M_{pz}^{(2)}$$

$$\frac{M_{xxz}}{2} = M_{pz}^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & M_{yzy} & 0 \\ -M_{yzy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzy} = M_{pz}^{(3)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} M_{xxxx} & 0 & M_{xxzz} & 0 & 0 & 0 \\ 0 & -M_{xxxx} & -M_{xxzz} & 0 & 0 & 0 \\ M_{xxzz} & -M_{xxzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & -M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxx} = 2M_{dv}^{(1)} + 4M_{dv}^{(2)} + M_{gv}^{(1)}$$

$$M_{xxzz} = M_{dv}^{(1)} - M_{gv}^{(1)}$$

$$M_{yzyz} = -M_{dv}^{(2)} + M_{gv}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & M_{xxyy} & M_{xxzz} & 0 & 0 & 0 \\ -M_{xxyy} & 0 & -M_{xxzz} & 0 & 0 & 0 \\ -M_{xxzz} & M_{xxzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxyy} = -2M_{dv}^{(3)} + 4T_{f3}^{(1)}$$

$$M_{xxzz} = -M_{dv}^{(3)} - 4T_{f3}^{(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} M_{yzyz} & 0 & 0 \\ 0 & -M_{yzyz} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzyz} = 2M_{dv}^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzyz} & 0 & 0 \\ 0 & M_{yzyz} & 0 \\ 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{yzyz} = -M_{dv}^{(5)} + T_{f3}^{(2)}$$

$$M_{xyxy} = 2M_{dv}^{(5)} + T_{f3}^{(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{yzyz} = -M_{dv}^{(6)} + T_{f3}^{(3)}$$

$$M_{xyxy} = 2M_{dv}^{(6)} + T_{f3}^{(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} M_{xxxx} & M_{xxyy} & M_{xxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xxyy} & -M_{xxxx} & -M_{xxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ M_{zzxx} & -M_{zzxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{yzyz} & 0 & 0 & M_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & -M_{yzzy} & 0 & 0 & -M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{xyxy} & 0 & 0 & -M_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
M_{xxxx} &= 2M_{dv}^{(1)} + 4M_{dv}^{(2)} + M_{gv}^{(1)} \\
M_{xxyy} &= -2M_{dv}^{(3)} + 4T_{f3}^{(1)} \\
M_{xxzz} &= M_{dv}^{(1)} - M_{dv}^{(3)} - M_{gv}^{(1)} - 4T_{f3}^{(1)} \\
M_{zzxx} &= M_{dv}^{(1)} + M_{dv}^{(3)} - M_{gv}^{(1)} + 4T_{f3}^{(1)} \\
M_{yzyz} &= -M_{dv}^{(2)} - M_{dv}^{(5)} + M_{gv}^{(1)} + T_{f3}^{(2)} \\
M_{yzyy} &= -M_{dv}^{(2)} + M_{dv}^{(5)} + M_{gv}^{(1)} - T_{f3}^{(2)} \\
M_{xyxy} &= 2M_{dv}^{(5)} + T_{f3}^{(2)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix}
0 & 0 & 0 & M_{yzyz} & 0 & 0 & M_{yzyy} & 0 & 0 \\
0 & 0 & 0 & 0 & M_{yzyy} & 0 & 0 & M_{yzyz} & 0 \\
0 & 0 & 0 & 0 & 0 & M_{xyxy} & 0 & 0 & M_{xyxy}
\end{bmatrix}$$

$$\begin{aligned}
M_{yzyz} &= 2M_{dv}^{(4)} - M_{dv}^{(6)} + T_{f3}^{(3)} \\
M_{yzyy} &= -2M_{dv}^{(4)} - M_{dv}^{(6)} + T_{f3}^{(3)} \\
M_{xyxy} &= 2M_{dv}^{(6)} + T_{f3}^{(3)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix}
M_{xxxx} & 0 & 0 \\
0 & -M_{xxxx} & 0 \\
0 & 0 & 0 \\
0 & 0 & M_{yyzz} \\
M_{zzxx} & 0 & 0 \\
0 & -M_{yyzz} - M_{zzxx} & 0 \\
0 & M_{yzyy} & 0 \\
0 & 0 & M_{zzxz} \\
-M_{yzyy} - M_{zzxz} & 0 & 0 \\
0 & 0 & 0
\end{bmatrix}$$

$$\begin{aligned}
M_{xxxx} &= 2M_{dv}^{(1)} + 4M_{dv}^{(2)} + M_{gv}^{(1)} \\
M_{yyzz} &= -M_{dv}^{(1)} + M_{dv}^{(3)} + M_{gv}^{(1)} \\
M_{zzxx} &= M_{dv}^{(1)} + M_{dv}^{(3)} - M_{gv}^{(1)} \\
M_{yzyy} &= -M_{dv}^{(2)} + M_{dv}^{(5)} + M_{gv}^{(1)} \\
M_{zzxz} &= M_{dv}^{(2)} + M_{dv}^{(5)} - M_{gv}^{(1)}
\end{aligned}$$